Heat transfer and getting back to Python
Modelling 2025

Plan for today:

1. Practice formulating models
2. Learn more about heat transfer
3. Get back to Python

See also:
Rasmuson Section 3.2,
Incropera 1.1 & 1.2,
terms.md (transport part),
py_data.md
Coffee, tank, and ice exercises

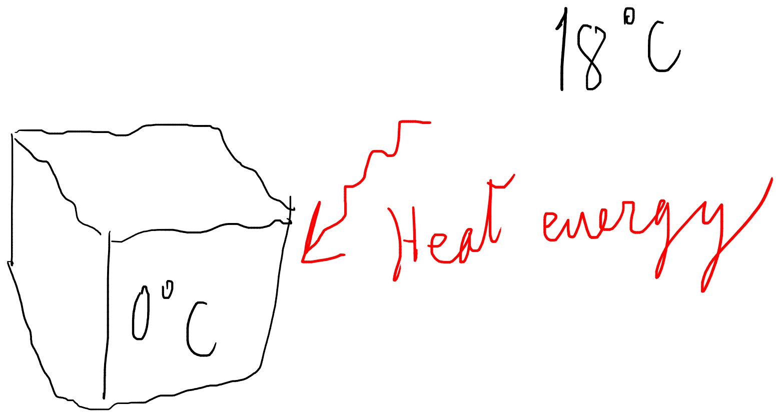
12.15-12.20: Goat warm-up problem
12.20-12.45: Tank exercise (simple model formulation)
12.45-13.00: Heat transfer lecture
13.00-13.10: Break
13.10-13.30: Heat transfer lecture
13.30-13.40: Menti quiz
13.40-14.00: Coffee exercise or other questions

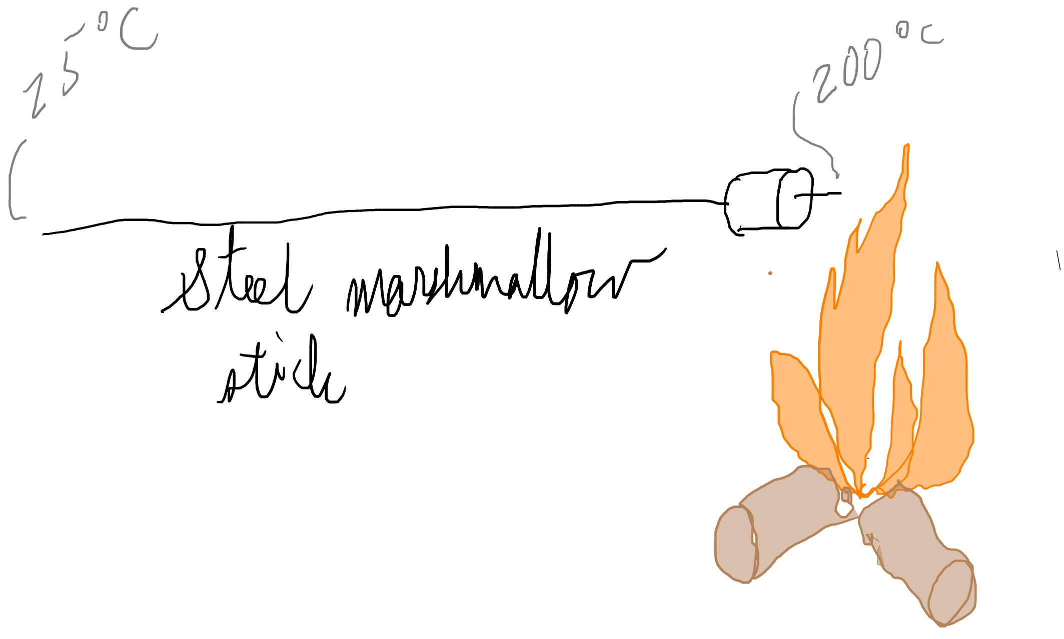
Start with warm-up problem 2 on goats and mountain lions

menti.com
4365 1979

Heat flow

Heat flows down temperature gradients, from a warmer location to a cooler one





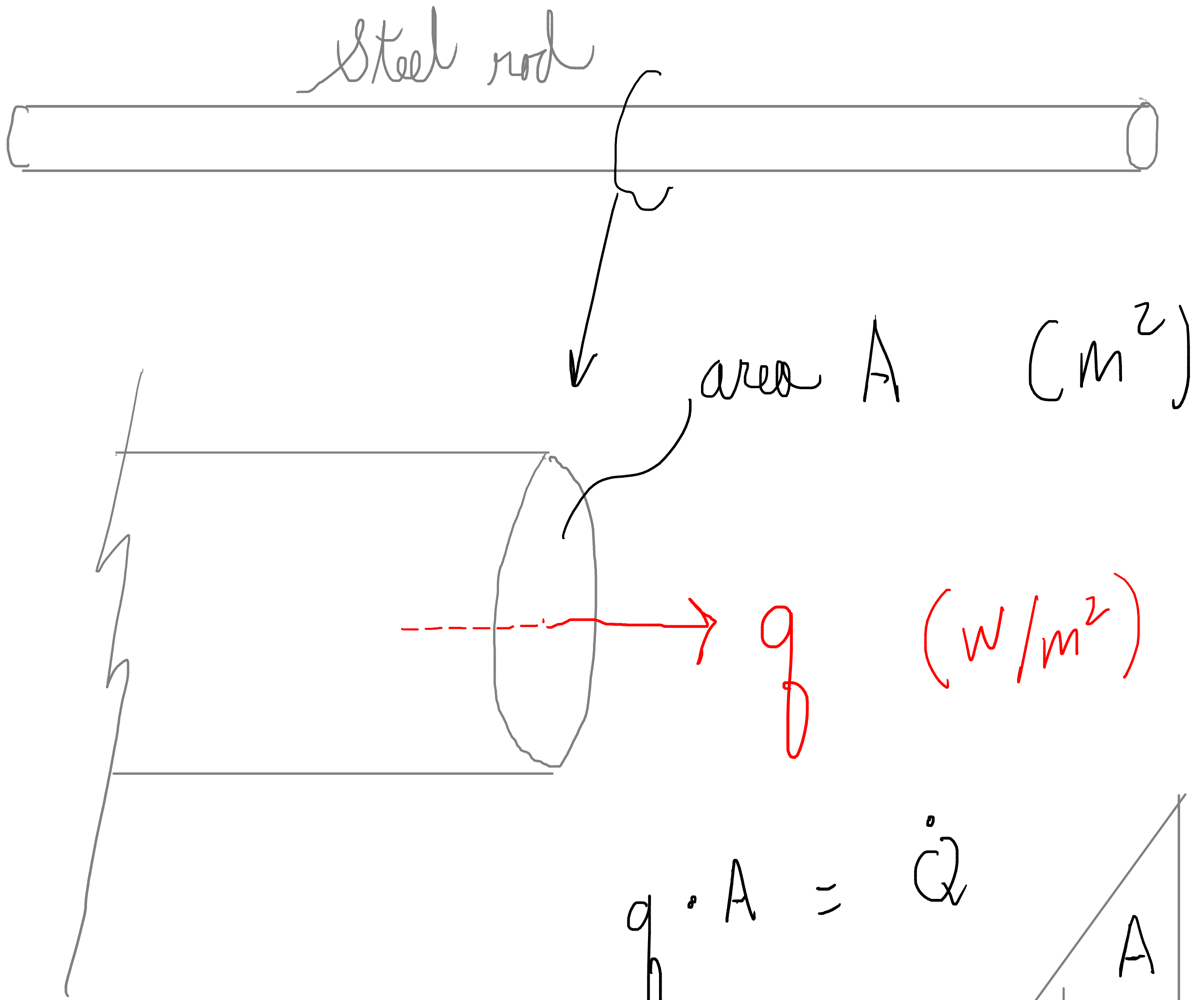
Heat energy flow and flux

Heat is energy, and heat flow typically has units of $\mathrm{J\ s^{-1} \equiv W}$

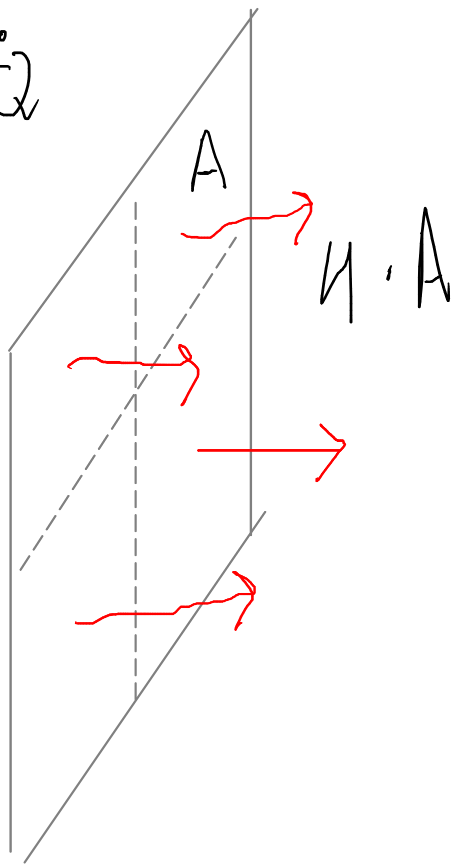
$$\dot{Q}$$

But it is heat flux that fundamentally depends on temperature gradients and material properties

$$q$$



$$q \cdot A = \dot{Q}$$



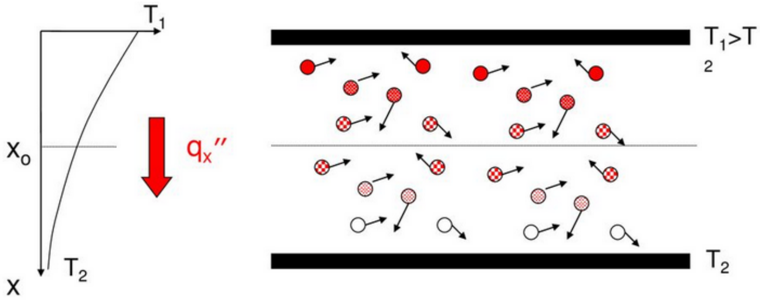
Typical heat flux units $\mathrm{J\ s^{-1} m^{-2} \equiv W m^{-2}}$

Heat transfer modes and mechanisms

Heat transfer is classified into modes: conduction, convection, radiation, advection

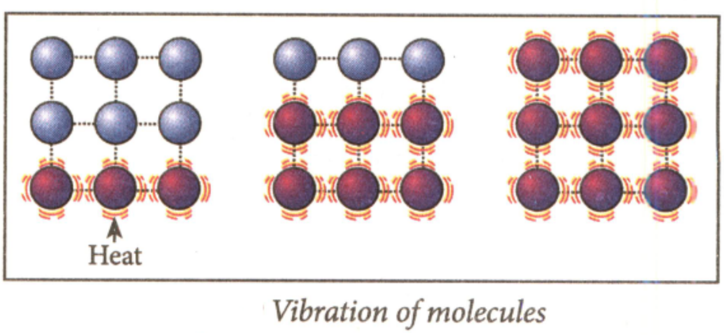
Each has an associated molecular mechanism, but the laws and parameters are phenomenological (especially convection)

Conduction: diffusion of heat (random molecular motion)



In fluids, transport of molecules

Vermeiren, https://slideplayer.com/slide/15519008/



Vibration of molecules

In solids, includes vibrational motion or transport of electrons (metals)

https://www.embibe.com/questions/Explain-conduction-using-molecular-theory./EM9713738

Conduction

Fundamental rate law for conduction is Fourier's law:

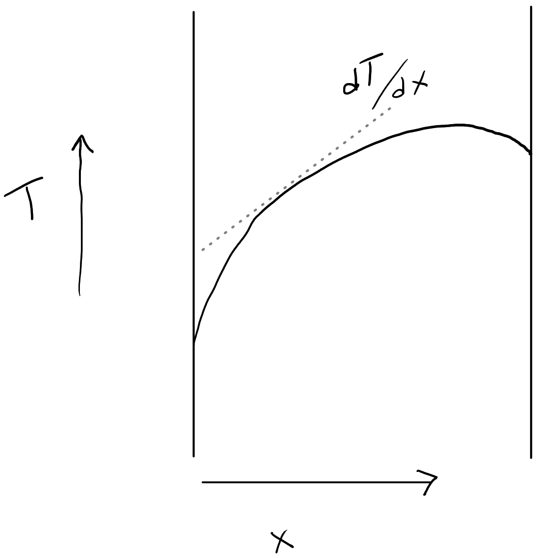
$$q_x = -k\frac{dT}{dx}$$

q_x

k

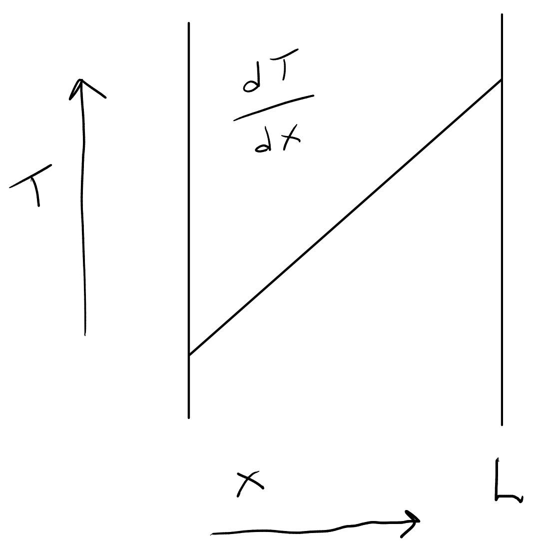
T

x



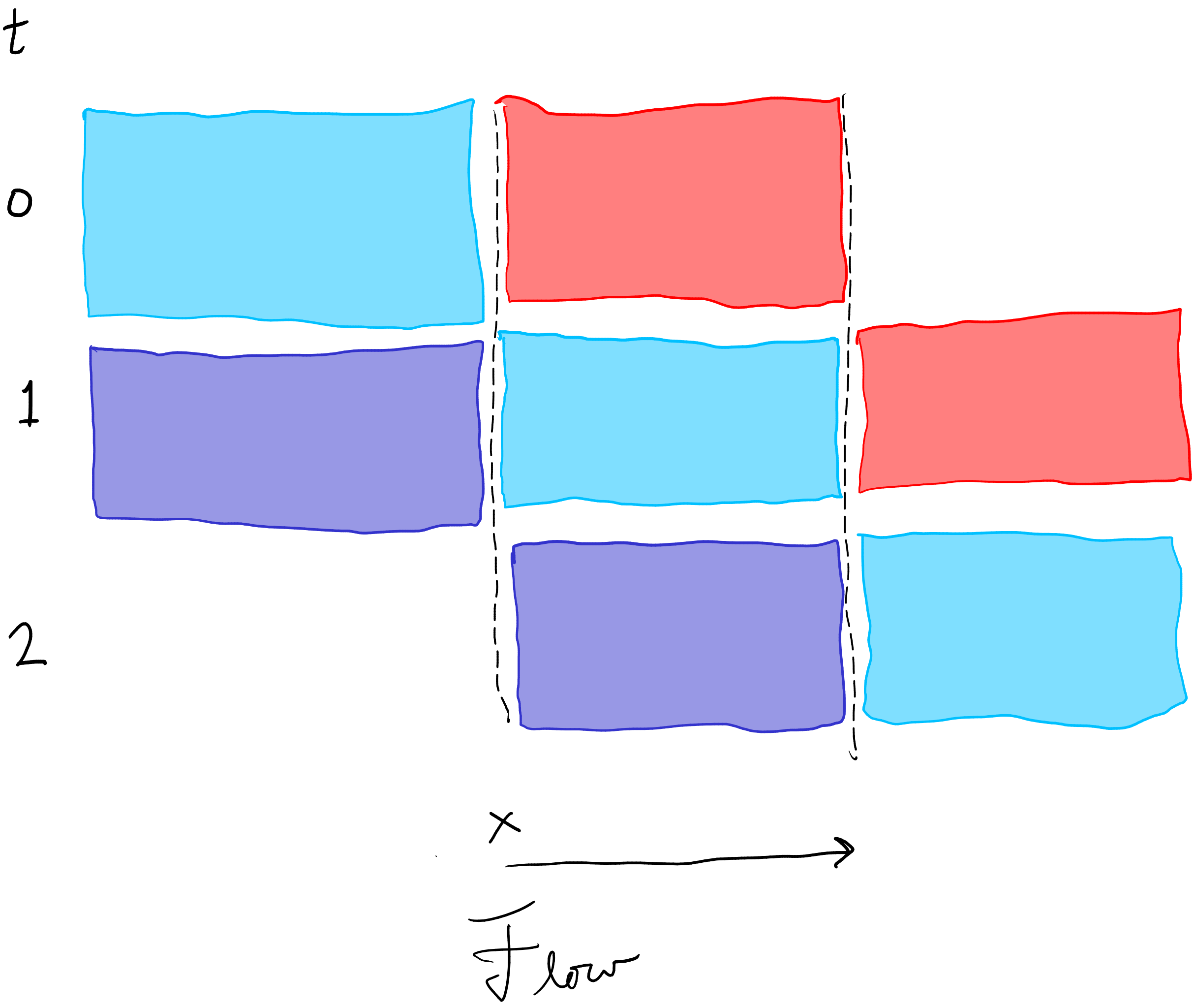
Steady-state conditions with linear temperature gradient:

$$q_x = -k\frac{T_2 - T_1}{L}$$



Advection

Think of parcels of fluid carrying thermal energy, displacing parcels with a different temperature



Rate of change in thermal energy is related to temperature gradient over space and fluid flow rate

$$-v_x \cdot c_p \cdot \frac{\partial T}{\partial x}$$

v_x

c_p

$$\frac{\partial T}{\partial t} = -v_x \frac{\partial T}{\partial x}$$

The heat equation

The governing equation for heat transfer that includes conduction and advection.
It describes what goes on within a model boundary.
It can be a starting point for model development for transport.

Rasmuson Appendix A.1

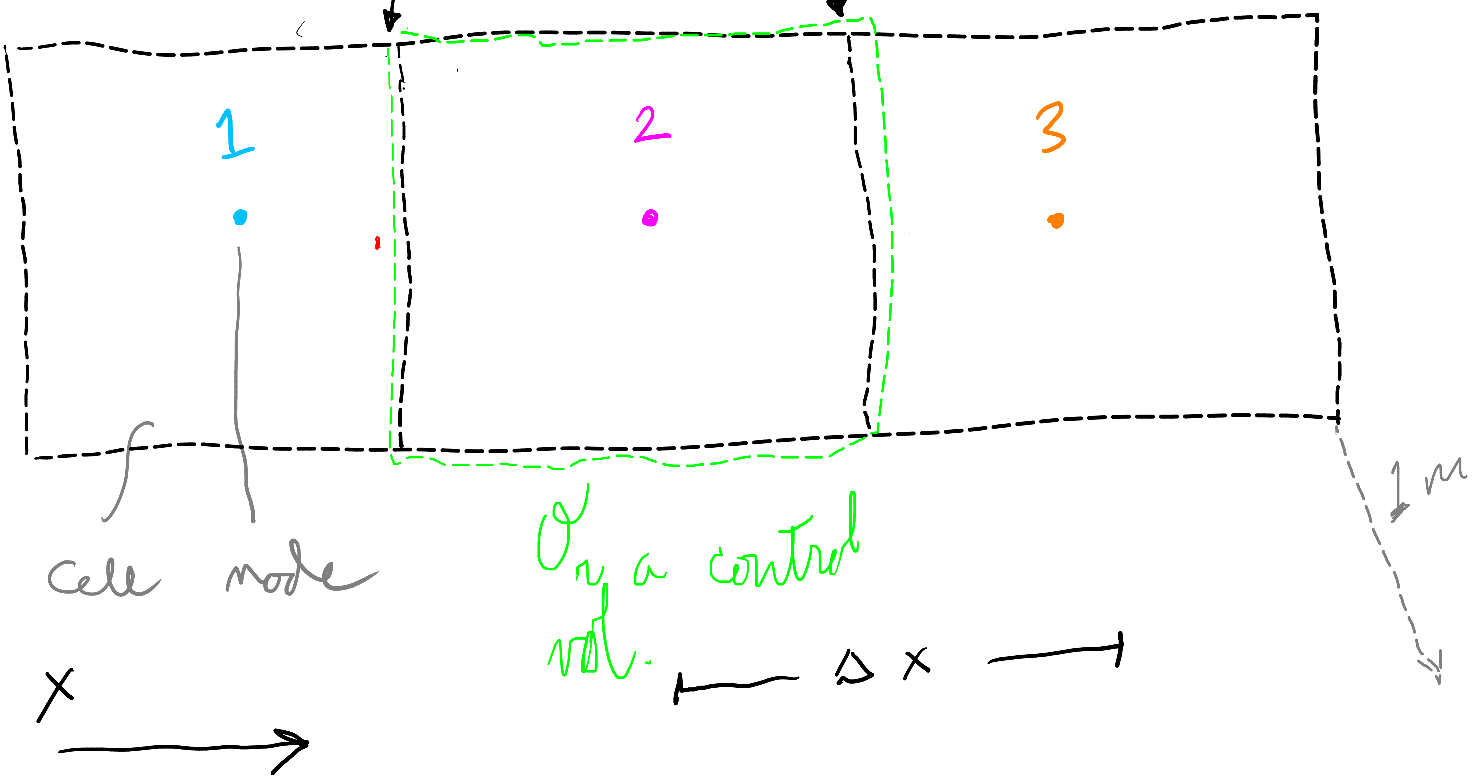
Rectangular coordinates

$$\rho c_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + S.$$

These equations can be derived from energy balance and rate equations.

Analogous difference (vs. differential) forms can help in understanding and model formulation

$$q_{x,in} \approx -k \frac{(T_2 - T_1)}{\Delta x} \qquad q_{x,out} \approx -k \frac{(T_3 - T_2)}{\Delta x}$$



$$\frac{dT_2}{dt} \approx \frac{q_{x,in} - q_{x,out}}{\rho c_p \Delta x}$$

$$= \frac{-k\left((T_2 - T_1) - (T_3 - T_2)\right)}{\rho c_p (\Delta x)^2}$$

Take limit as $\Delta x \to 0$

$$\frac{\partial T}{\partial t} = \frac{k}{\rho c_p} \cdot \frac{\partial^2 T}{\partial x^2}$$

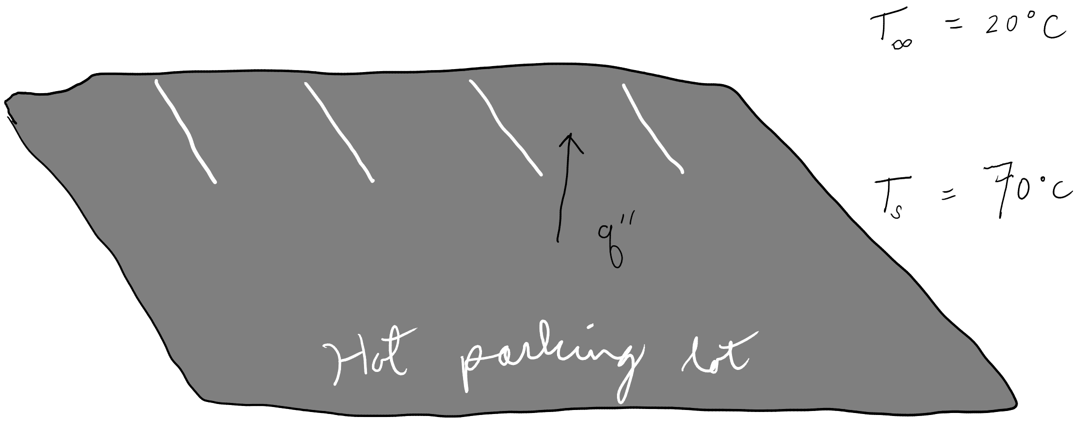
$$\rho c_p \left(\frac{\partial T}{\partial t} + \cancel{v_x \frac{\partial T}{\partial x}} + \cancel{v_y \frac{\partial T}{\partial y}} + \cancel{v_z \frac{\partial T}{\partial z}} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \cancel{\frac{\partial^2 T}{\partial y^2}} + \cancel{\frac{\partial^2 T}{\partial z^2}} \right) + \cancel{S}.$$

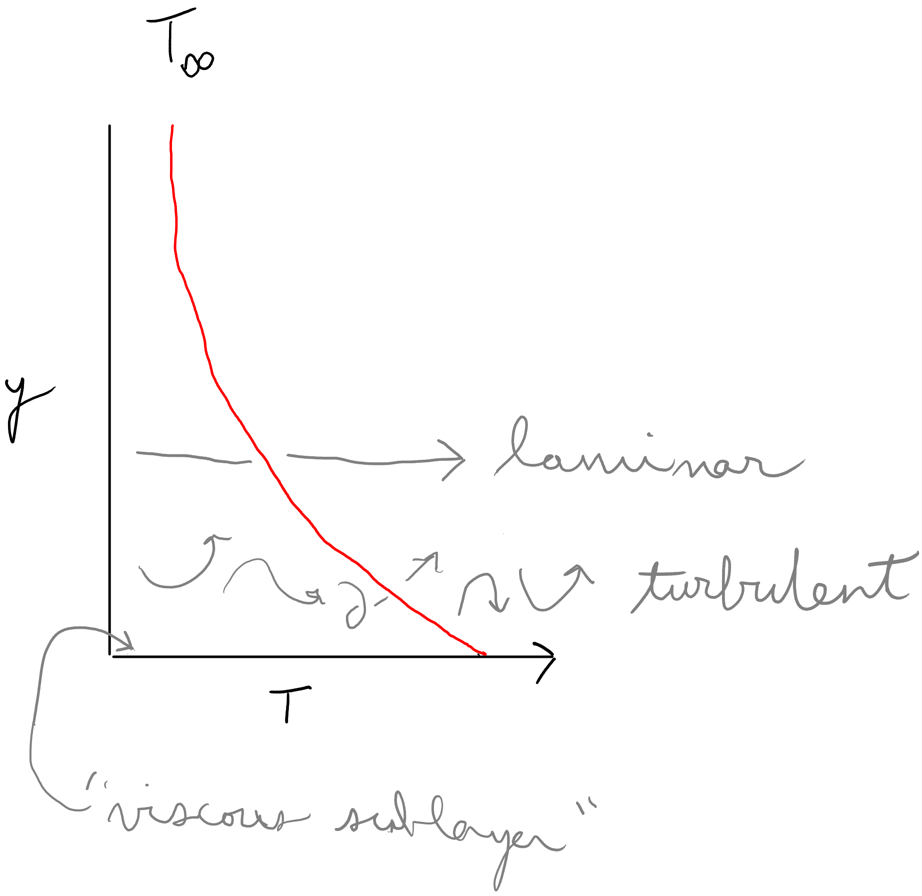
$$\frac{\partial T}{\partial t} = \frac{k}{\rho c_p} \frac{\partial^2 T}{\partial x^2}$$

More with the "numerical method of lines" (NMOL) in a couple weeks.

Convection - at the system boundary

Underlying mechanisms for convection include both random molecular and bulk motion





Newton's law of cooling is the fundamental constitutive equation for convection:

$$q = h(T_s - T_\infty)$$

q

T_s

T_∞

h

TABLE 1.1 Typical values of the convection heat transfer coefficient

Process	h ($W/m^2 \cdot K$)
Free convection	
Gases	2–25
Liquids	50–1000
Forced convection	
Gases	25–250
Liquids	100–20,000
Convection with phase change	
Boiling or condensation	2500–100,000

Incropera et al. (2003)

Heat transfer coefficient "correlations" are widely developed and used

The general form of Newton's law of cooling is a flexible and convenient tool for developing simple heat transfer models

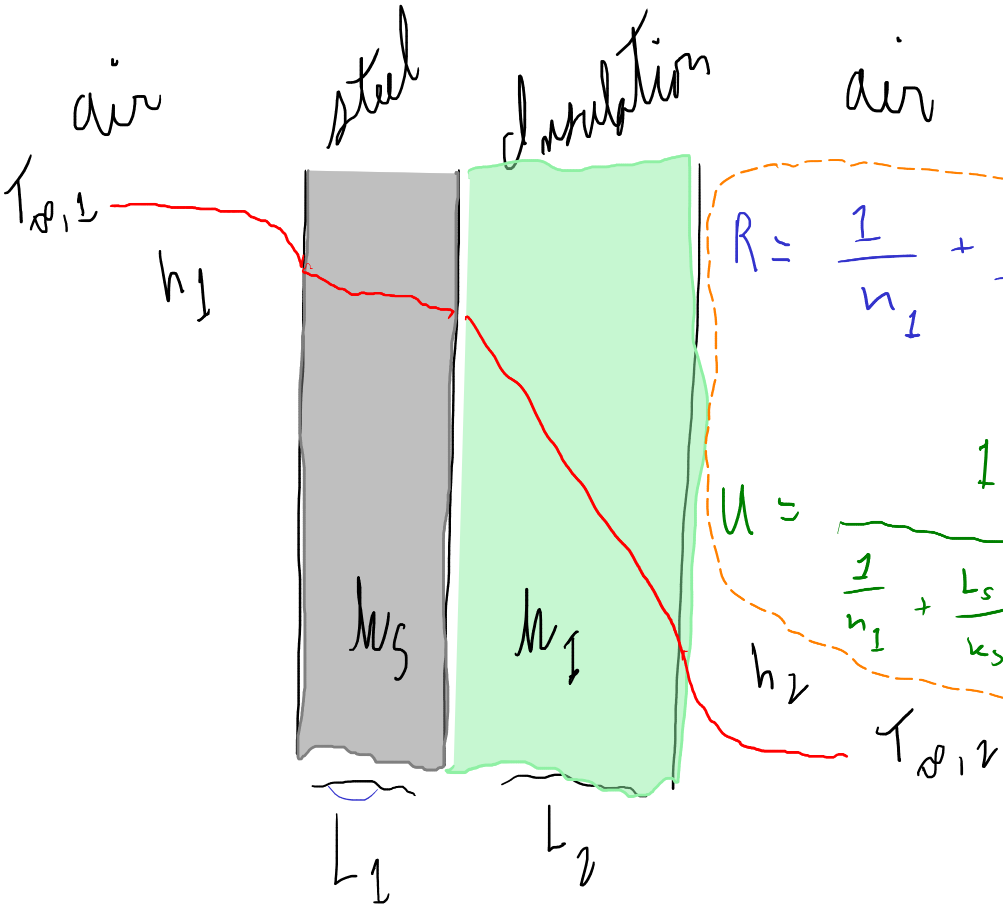
Thermal resistance and overall heat transfer coefficient

Heat transfer modes in series can be combined and expressed as an overall heat transfer coefficient or a resistance

$$q = U \cdot \Delta T$$

$U \equiv$

$$R = \frac{1}{U \cdot A} \quad \text{or} \quad R = \frac{1}{U}$$



$$R = \frac{1}{h_1} + \frac{L_S}{k_S} + \frac{L_I}{k_I} + \frac{1}{h_2}$$

$$U = \frac{1}{\frac{1}{h_1} + \frac{L_s}{k_s} + \frac{L_I}{k_I} + \frac{1}{h_2}}$$

Corrected
16 Sept. 2025

U or R can be calculated from individual layers or may be completely empirical

And these parameters may include other modes of transport

Radiation

Radiation is thermal energy transmitted by electromagnetic waves or photons

$$q_{rad} = \epsilon\sigma(T_s^4 - T_{sur}^4)$$

Radiation becomes important when temperature differences are high.

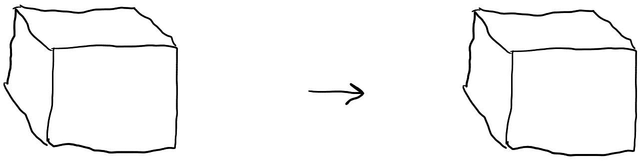
We won't do much with radiation in this course--wait for the transport phenomena course!

Sensible and latent heat

Another complication in heat transfer comes from phase changes

Sensible energy is reflected by a change in the temperature of a material (see specific heat)

Latent energy has to do with phase changes: freezing, melting, vaporization, condensation, boiling



0° C → 10° C

Liquid → Liquid

Solid → Liquid

Major effects on thermal energy! Why?

TABLE A.6 Thermophysical Properties of Saturated Water

Temperature, T (K)	Pressure, p (bars)[b]	Specific Volume (m³/kg) $v_f \cdot 10^3$	Specific Volume (m³/kg) v_g	Heat of Vaporization, h_{fg} (kJ/kg)	Specific Heat (kJ/kg · K) $c_{p,f}$	Specific Heat (kJ/kg · K) $c_{p,g}$
273.15	0.00611	1.000	206.3	2502	4.217	1.854
275	0.00697	1.000	181.7	2497	4.211	1.855
280	0.00990	1.000	130.4	2485	4.198	1.858
285	0.01387	1.000	99.4	2473	4.189	1.861
290	0.01917	1.001	69.7	2461	4.184	1.864

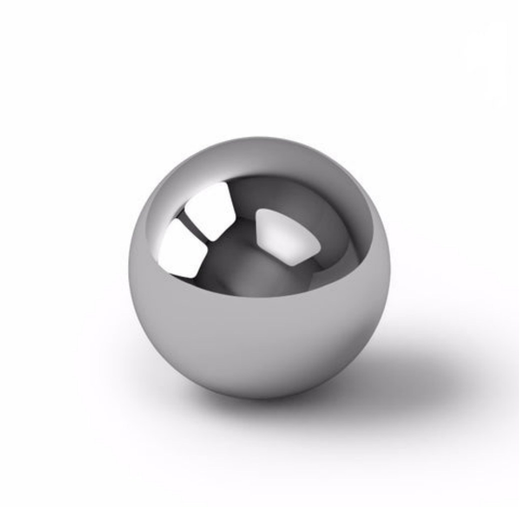
Substance	Heat of fusion (cal/g)	Heat of fusion (J/g)	Melting temperature, °C
water	79.72	333.55	0.0
methane	13.96	58.99	−182.46
propane	19.11	79.96	−187.7
glycerol	47.95	200.62	17.8
formic acid	66.05	276.35	8.4
acetic acid	45.90	192.09	16 - 17

Simplifying and solving heat transfer models

It is difficult to start with the complete heat equation and develop an analytical model without some simplifications.

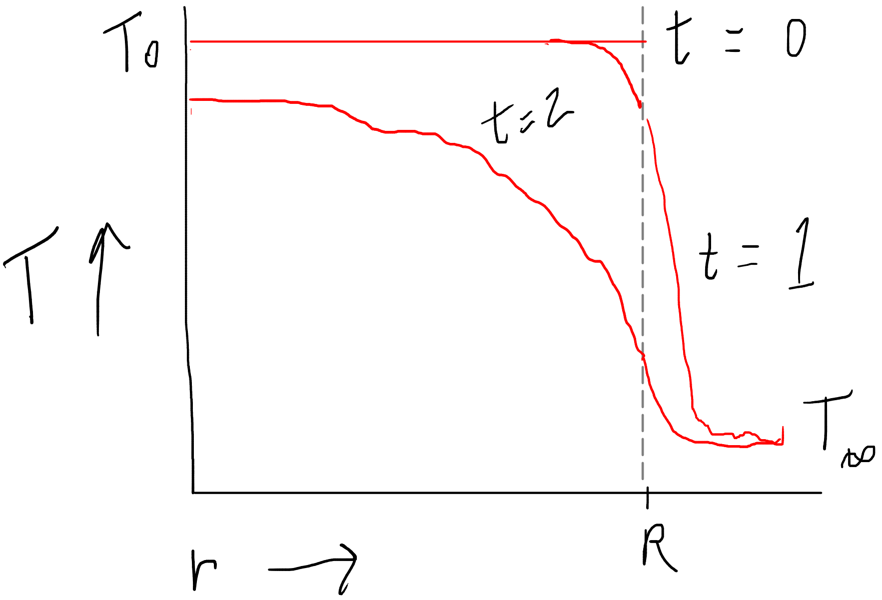
Near the start of a model formulation process, think about whether any of these simplifications can be applied:
1. Lumped capacitance (lumped parameter),
2. steady-state,
3. simplified geometry

An example: A hot steel ball (1 cm diameter, 400 °C) cools in ambient air.
How long does it take?



With the lumped capacitance approach we ignore internal resistance

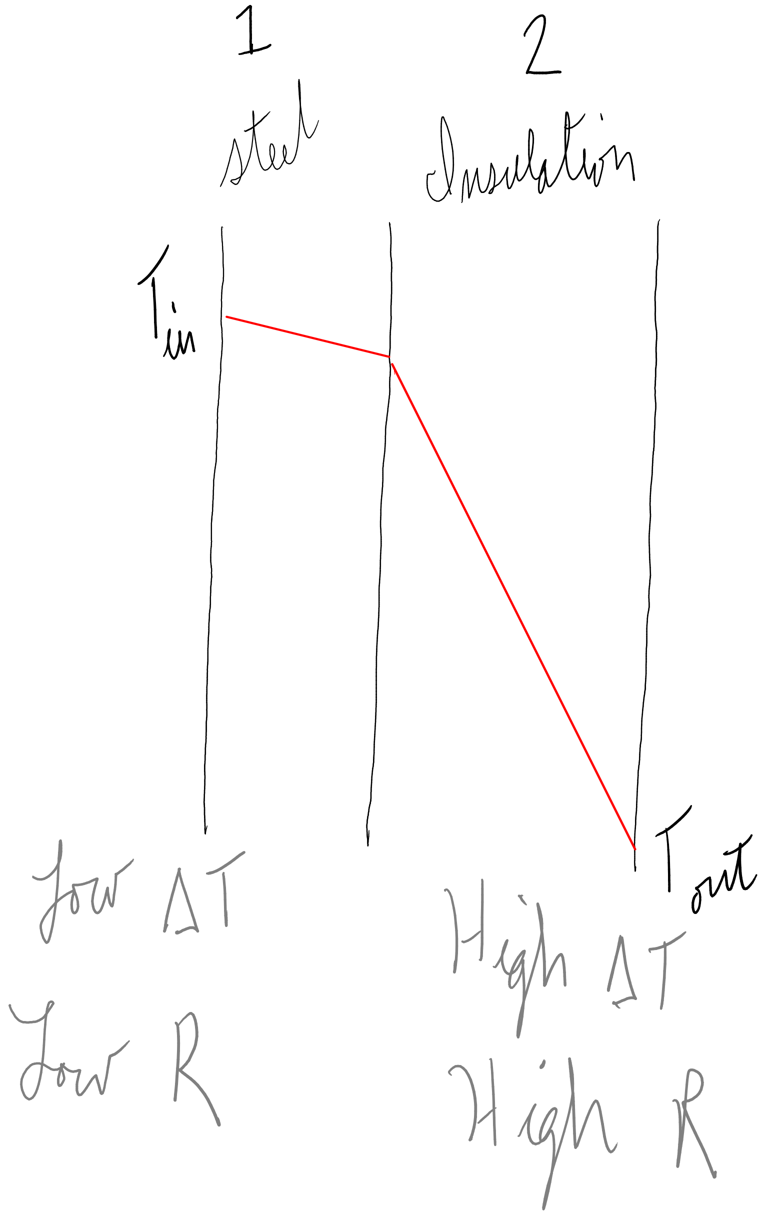
Let's start with evaluation of a lumped capacitance approach. Is it appropriate?



Comparing internal and external resistance

The key to appropriate use of lumped capacitance approach is to compare internal resistance to external resistance.

Steady-state temperature difference and heat transfer resistance are inversely related.



If $R_2 >> R_1$ then $R_2 + R_1 \approx R_2$

The Biot number

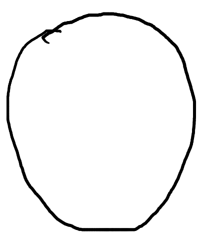
Bi = Biot number = approximate ratio of internal : external resistance

$$Bi = \frac{h}{k} L$$

characteristic length.
viz. $\frac{V}{A}$ or farthest conduction distance

$Bi < 0.1 \rightarrow$ lumped capacitance OK!

$Bi > 0.1 \rightarrow$ must consider internal gradients.



1 cm ball

$$Bi = \frac{10\ W m^{-2} K^{-1} \cdot 0.0033 m}{50\ W m^{-1} K^{-1}} = 0.00067$$

So go ahead and lump!

$$\frac{V}{A} = \frac{4/3 \pi r^3}{4 \pi r^2} = \frac{r}{3} = 0.0033 m$$

k for steel $\approx$ 50 $W m^{-1} K^{-1}$ but it varies.

Quizzes :)

Python data structures: **menti.com 1895 6212**

Heat transfer: **menti.com 4679 7160**